

VOA

data:

- A vector space V
- $\Omega \in V$ vacuum vector
- $T: V \rightarrow V$ translation
- a linear map $Y: V \rightarrow \text{End}(V)[[z, z^{-1}]]$ state-field correspondence
- $\omega \in V$ conformal vector

$V[[a, b]] / (ab - 1)$

Axioms:

- * $Y(a, z)b$ this is a power series in z with coefficients in V . it is bounded below.

* Identity

$$Y(\Omega, z)u = u$$

$$Y(u, z)\Omega = u + \mathcal{O}(z)$$

* Translation

$$T\Omega = 0$$

$$[T, Y(a, z)]b = \frac{d}{dz} Y(a, z)b$$

* Locality

$$\forall a, b, \exists N \gg 0$$

$$\text{s.t. } (z_1 - z_2)^N [Y(a, z_1), Y(b, z_2)] = 0$$

$$* Y(\omega, z) = \sum_{n=-\infty}^{\infty} L_n z^{-n-2} \quad \text{where } L_n \in \text{End}(V) \quad \forall n \in \mathbb{Z}$$

$$(i) [L_m, L_n] = (m-n)L_{m+n} + \frac{c}{12} \delta_{m+n,0} (m+1)m(m-1) \text{id}_V$$

(ii) L_0 acts semi-simply on V with integer eigenvalues bounded below.

(iii) let V_n be the eigenspace of L_0 corresponding to $n \in \mathbb{Z}$

$$V = \bigoplus_{n=-\infty}^{\infty} V_n \quad (\text{gives grading})$$

let $a \in V_m$ $b \in V_n$

then the coefficient of z^k in $Y(a, z)b \in V_{n+m+k}$

$$(iv) \quad \Omega \in V_0, \quad \omega \in V_2$$

$$(v) \quad L_{-1} = T$$

Unitary

data

$$\langle, \rangle : V \times V \longrightarrow \mathbb{C}$$

inner product

$$\Theta : V \longrightarrow V$$

involution
(anti-linear VOA automorphism)
fixes Ω, c_0 .

Axiom

$$\star \quad \langle a, Y(b, z)u \rangle = \langle Y(b, z)^{\dagger} a, b \rangle \quad (-z^{-2})^n b \text{ when } b \in V_n$$

$$\text{where } Y(b, z)^{\dagger} := Y(\Theta e^{zL_1} \overline{(-z^{-2})}^{L_0} b, \bar{z}^{-1})$$

for $a_1, \dots, a_n \in V$ $\star \quad V = \bigoplus_{n=0}^{\infty} V_n$ [i.e. eigenvalues of L_0 are non-negative]

power series of n -variables $\hat{V} = \prod_{n=0}^{\infty} V_n$ $\hat{V}$ valued function on $\mathbb{C}^n \setminus \Delta := \{z_i \in \mathbb{C}^n : z_i \neq z_j\}$

$\nabla \quad \left[\begin{array}{l} Y(a_1, z_1) \dots Y(a_n, z_n) \Omega \\ \equiv a_1(z_1) \dots a_n(z_n) \Omega \end{array} \right] \xrightarrow{\text{converges}} \hat{V}$ for $|z_1| > \dots > |z_n| \geq 0 \rightarrow$ analytic continuation.

$\hat{V}$ valued function on $\mathbb{C}^n \setminus \Delta$ $\xrightarrow{\text{extension}}$

Conformal Net

data: $\bullet$ Hilbert space $\mathcal{H}_0$, called the vacuum sector, $\Omega \in \mathcal{H}_0$, vacuum vector

$\bullet$ a projective representation of orientation preserving diffeomorphism group of the circle $U : \text{Diff}^+(S^1) \longrightarrow \text{PU}(\mathcal{H}_0)$ which is an honest representation of the Möbius subgroup (the automorphisms $S^1 \rightarrow S^1$ obtained from restriction to boundary of complex automorphisms of the unit disc)

$\bullet \quad \forall$ interval $I \subset S^1$, a von Neumann $A(I) \subseteq \mathcal{B}(\mathcal{H}_0)$
closed under WOT on $\mathcal{B}(\mathcal{H}_0)$

Axioms:

$\bullet$ If $I_1 \subset I_2$, $A(I_1) \subset A(I_2)$

$\bullet$ Ω is cyclic for the algebra generated by all $A(I)$, Ω is fixed under $\text{Möb} \subset \text{Diff}^+(S^1)$

$\bullet$ U is a positive energy representation (the spectrum of L_0 is positive and eigenspaces are finite dimensional)

$\bullet$ Covariance: $U(\phi) A(I) U(\phi)^{-1} = A(\phi(I))$

$$\phi|_I = \text{id}_I \Rightarrow \text{Ad}(U(\phi))|_{A(I)} = \text{id}_{A(I)}$$

• Locality : $\nexists \dot{I}_1 \cap \dot{I}_2 = \emptyset, [A(I_1), A(I_2)]_{BCH_0} = 0$

Consequence : • Haag duality $A(S' \setminus I) = A(I)'$ in $B(H_0)$

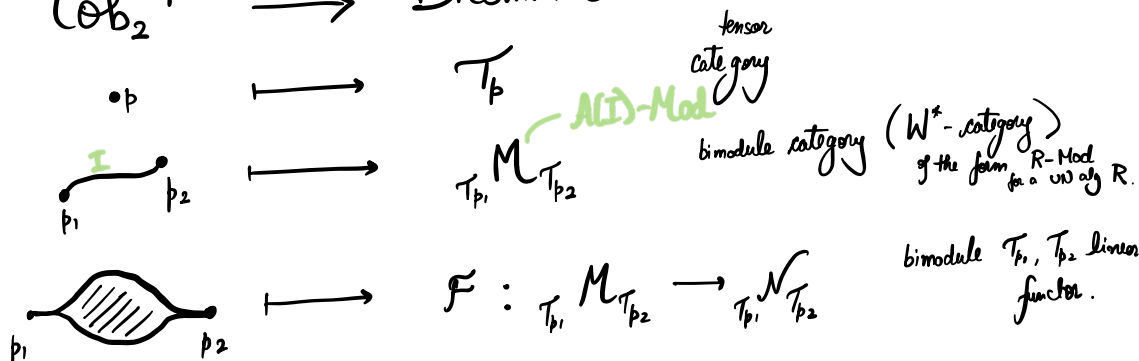
• Ω is cyclic for each $A(I)$ [Reeh - Schlieder]

• $A(I)$ makes sense for abstract interval I and produces an abstract von Neumann algebra.

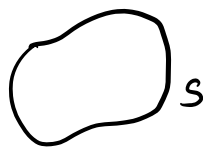
Segal X CFT (fully extended) unitary

$Z : \text{Cob}_2^{\text{conf}} \longrightarrow \text{Bicommod Cat}$

sym monoidal functor



defn.



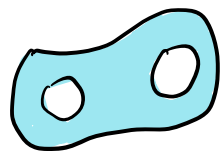
linear category $\mathcal{C}(S)$

$U : \mathcal{C}(S) \longrightarrow \text{Hilb}$
pointing



$M \begin{smallmatrix} \boxed{X} \\ \hline T_{p_2} \oplus T_{p_1} \end{smallmatrix} N$
 W^* -cat

End $U \rightsquigarrow$ alg of observables.



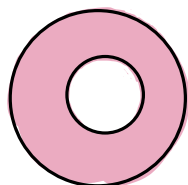
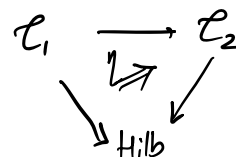
$F_\Sigma : \mathcal{C}(\partial_{\text{in}} \Sigma) \longrightarrow \mathcal{C}(\partial_{\text{out}} \Sigma)$

$F_\Sigma = H_\Sigma \otimes -$

$\eta_\Sigma = \Omega_\Sigma \otimes -$

and $\forall \lambda \in \mathcal{C}(\partial_{\text{in}} \Sigma)$

$\eta_\Sigma : U(\lambda) \longrightarrow U(F_\Sigma(\lambda))$



$\tilde{A} \in \tilde{\text{Ann}}$

$T_{\tilde{A}} : F_A \longrightarrow \text{id}_{\mathcal{C}(S)}$

$\forall \lambda \in \mathcal{C}(S),$

$\tilde{\text{Ann}}_{\tilde{A}}(S) \longrightarrow \text{End}(U(\lambda))$ is holomorphic.
 $\longmapsto U(T_{\tilde{A}}) \circ \eta_A$

VOA $\rightsquigarrow$ cNet

H_0 is completion of V w.r.t $\langle, \rangle$

cNet $\rightsquigarrow$ VOA

$\gamma = \bigoplus$ eigenspaces of rotations $\in \text{Diff}^+(S^1)$

unitary VOA

$\rightsquigarrow$ γ conformal nets

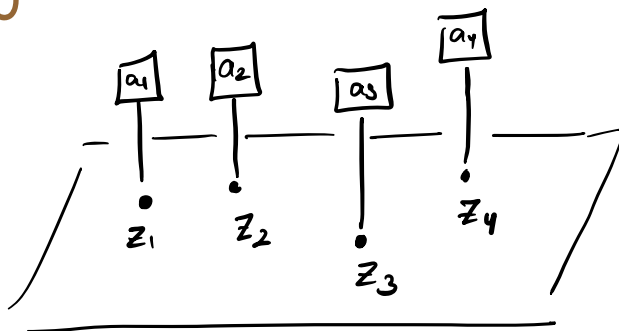
rational conformal nets
 $\{$
 finite index

$\rightsquigarrow$ unitary γ Segal CFT

Analog of $a_1(z_1) \dots a_n(z_n) \Omega \in \hat{V}$ in CNet

$$r_1^{L_0} a_1 \underbrace{\left(\frac{r_2}{r_1}\right)^{L_0} a_2 \left(\frac{r_3}{r_2}\right)^{L_0} \dots a_n}_{\text{bounded}} \Omega \in \hat{V}$$

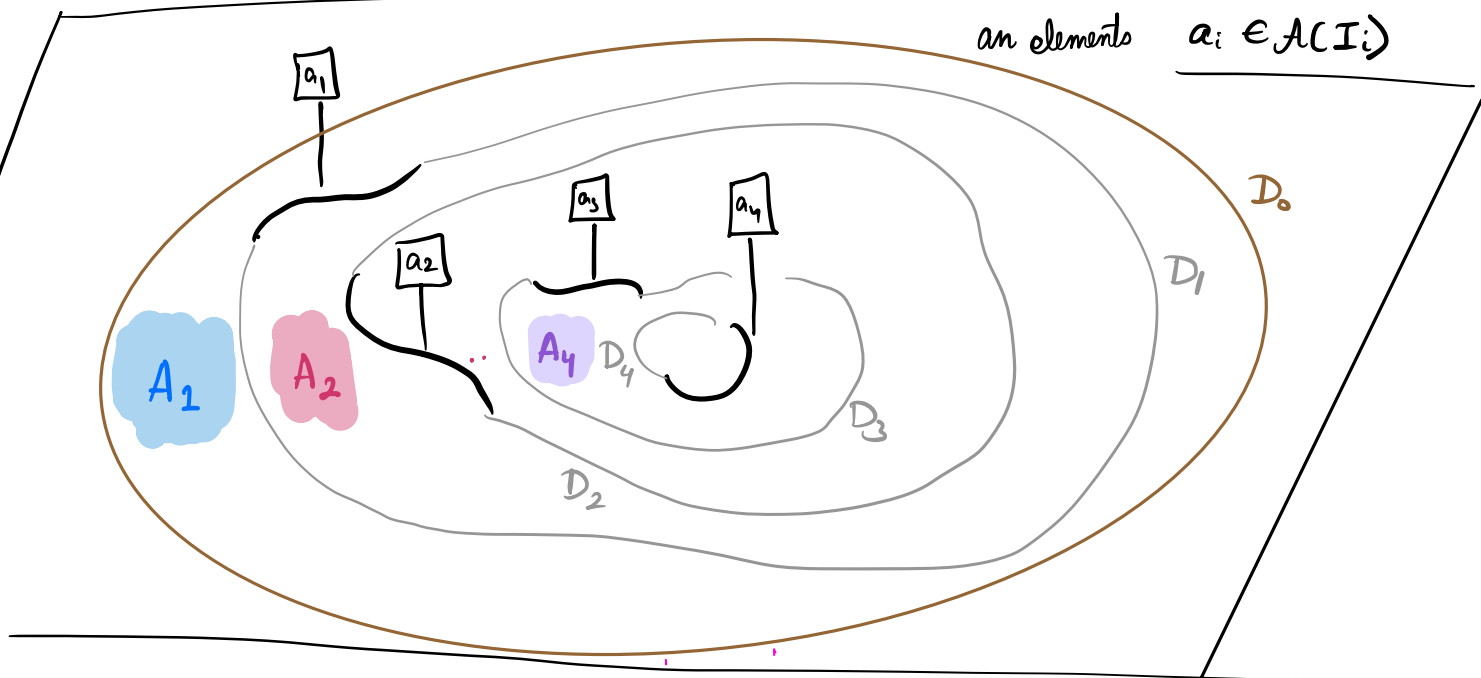
$\in H_0$



$\downarrow$
 $t_1 > t_2 > \dots > t_n$

given disjoint interval

$I_1, \dots, I_n \in \mathbb{C}$,
an elements $a_i \in A(I_i)$



$D_1 \supset D_2 \supset \dots \supset D_n$ discs in $\mathbb{C}$

$I_i \in \partial D_i$, $\varphi_i : \mathbb{D} \xrightarrow{\sim} D_i$

annuli $A_1 \dots A_n$

$$R^{L_0} \underline{A_1} \varphi_1^{-1}(a_1) \underline{A_2} \varphi_2^{-1}(a_2) \dots \underline{A_n} \varphi_n^{-1}(a_n) \Omega \in \hat{V}$$

operators

$$\text{Ann} = \{ (A, \varphi_{\text{in}}, \varphi_{\text{out}}) \mid A = \text{smooth closed curves.} \quad \varphi_{\text{in/out}} : S^1 \xrightarrow{\sim} \partial_{\text{in/out}} A \}$$

Monoid
Semi-group law
 associated to the
 complexification of the Lie algebra of $\text{Diff}(S^1)$
 $\mathcal{K} \otimes_{\mathbb{R}} \mathbb{C}$ with

A, A' in Ann

$$A \cup A' := A \sqcup A' \quad / \varphi_{\text{in}}(z) \sim \varphi'_{\text{out}}(z) \quad \text{gluing}$$

Sub-semigroup of
univalent map

$$\text{Univ} = \left\{ A \in \text{Ann} \mid A = \mathbb{D} \setminus f(\mathbb{D}) \right. \\ \left. \text{for } f: \mathbb{D} \rightarrow \mathbb{D} \text{ holomorphic embedding derivative } \neq 0 \text{ everywhere.} \right\}$$

$\varphi_{\text{in}} = f|_{\partial\mathbb{D}}, \varphi_{\text{out}} = \text{id}|_{\partial\mathbb{D}}$

at algebra level is
 $\text{Span} \{ L_n \}_{n \geq -1}$

Th: A unitary pos. en. rep. of Vir on $\mathcal{H}$ integrates to a holomorphic rep of $\tilde{\text{Ann}} \rightarrow \mathcal{B}(\mathcal{H})$

Virasoro cocycle

vanishes on $\text{Span} \{ L_n \}_{n \geq -1}$

$$\frac{c}{12} m(m-1)(m+1) \delta_{n+m,0} \text{ id}_{\mathcal{H}}$$

$$\text{central extension } \mathbb{Z} \times \mathbb{C}^x \longrightarrow \tilde{\text{Ann}} \xrightarrow{\text{Vir}} \text{Ann} \quad \text{with}$$

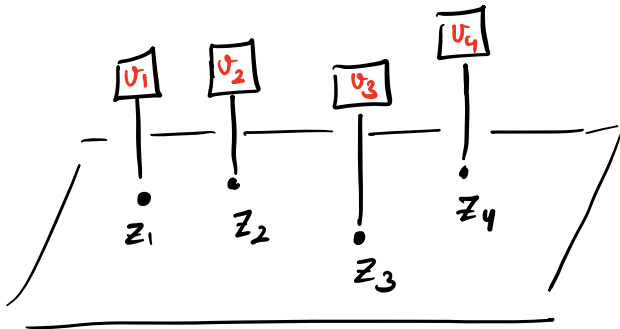
splits over $\text{Univ} \subset \text{Ann}$
 $\mathbb{C}$ central charge for rational CFTs $c \in \mathbb{Q}$

then Ann are like $\text{Diff}^+(S^1)$

in Univ (act by honest operators on the Hilbert space)

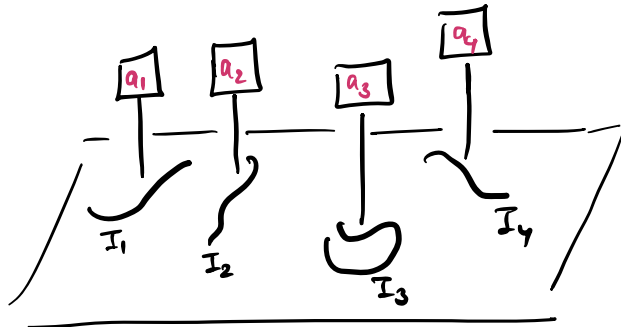
$$R^{L_0} \underbrace{A_1}_{\text{operators}} \varphi_1^{-1}(a_1) \underbrace{A_2}_{\text{operators}} \varphi_2^{-1}(a_2) \dots \underbrace{A_n}_{\text{operators}} \varphi_n^{-1}(a_n) \Omega \in \hat{\mathcal{V}}$$

Span { elements of $\hat{V}$ that can be expressed as field insertion } $\subset$ Span { elements of $\hat{V}$ that can be expressed as worm insertion } — lemma.



$$v_1, v_2, \dots, v_n \in V$$

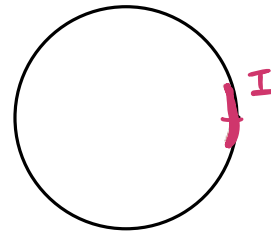
$$z_1, \dots, z_n \in \mathbb{C}, z_i \neq z_j$$



$$I_1, \dots, I_n \in \mathbb{C} \text{ disjoint}$$

$$a_i \in A(I_i)$$

Proof: case of a single field insertion at 0.
Fix $v \in V_n$ and $I \subset S^1$



By Reeh-Schlieder

$$A(I) \xrightarrow[\text{dense}]{\text{dense}} \mathcal{H} \subset \hat{V} \xrightarrow[\text{dense}]{} V_n$$

$a \mapsto a\Omega$
 $\varphi \mapsto \int_{S^1} z^{L_0-n-1} \varphi \frac{dz}{2\pi i}$

↖ finite dimensional

$$\text{projection} = \begin{cases} \varphi & \text{for } \varphi \in V_n \\ 0 & \text{otherwise} \end{cases}$$

ie

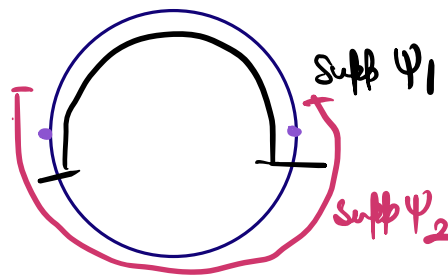
$$\exists \underline{a} \in \underline{A(I)} \text{ s.t. } v = \int_{S^1} z^{L_0-n-1} \underline{a} \Omega \frac{dz}{2\pi i}$$

Pick a partition of unit

$$1 = \psi_1 + \psi_2$$

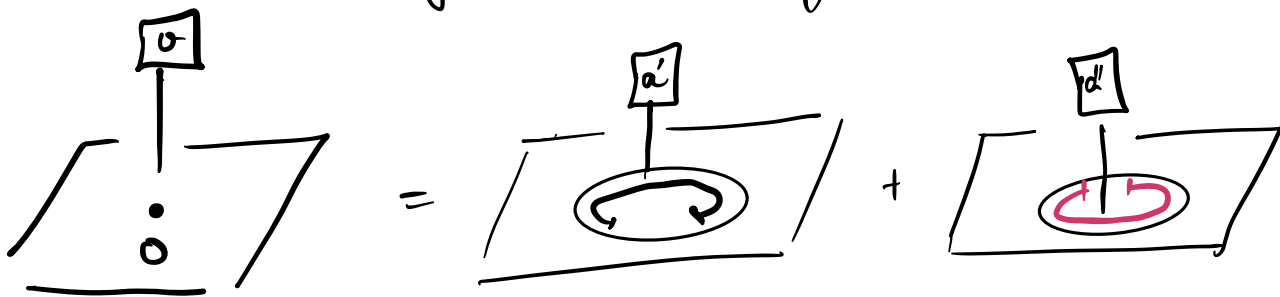
$$z = e^{i\theta}$$

$$R_0 = e^{i\theta_0}$$



$$\psi = \left(\underbrace{\left[\int_{S'_+} \psi_1(z) z^{L_0-n-1} a \frac{dz}{2\pi i} \right]}_{a'} + \underbrace{\left[\int_{S'_-} \psi_2(z) z^{L_0-n-1} a \frac{dz}{2\pi i} \right]}_{a''} \right) \Omega$$

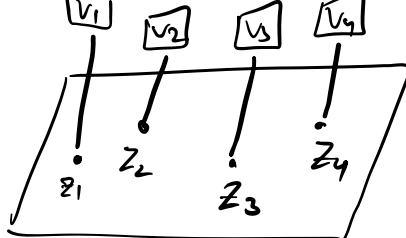
$\therefore$ we have written our field as a sum of two worm



every point insertion is a linear combination of two worm insertions.

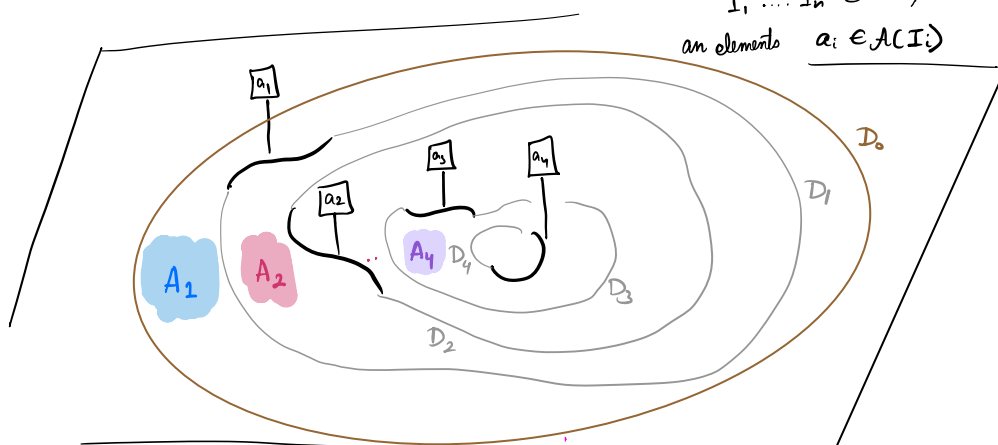
General case.

$$v_1(z_1) \cdots v_n(z_n) \Omega$$



$$= \sum_{2^n \text{ terms}} \left[\text{Diagram of } \Omega \text{ with four points } a_1', a_2'', a_3', a_n'' \text{ on the boundary, each enclosed in a red circle with a cross} \right] \Omega$$

use as defn of left hand side



given disjoint intervals
 $I_1, \dots, I_n \in \mathbb{C}$,
 an element $a_i \in A(I_i)$

$D_1 \supset D_2 \supset \cdots \supset D_n$ discs in $\mathbb{C}$
 $I_i \in \partial D_i$, $\varphi_i : \mathbb{D} \xrightarrow{\sim} D_i$

$$R^{\perp} \underbrace{A_1 \varphi_1^{-1}(a_1) A_2 \varphi_2^{-1}(a_2) \cdots A_n \varphi_n^{-1}(a_n)}_{\substack{\text{operators} \\ \in H}} \Omega \quad \begin{matrix} \text{annuli } A_1 \cdots A_n \\ e \in \hat{V} \end{matrix}$$

worms in $\mathbb{C} \rightsquigarrow$ in $\hat{V}$
 worms in $\mathbb{D} \rightsquigarrow$ in $\mathbb{H}$
 worms in $A \rightsquigarrow$ bounded linear map $\mathbb{H} \rightarrow \mathbb{H}$



fields in $\mathbb{C} \rightsquigarrow$ no $\hat{V}$
 fields in $\mathbb{D} \rightsquigarrow$ $\mathbb{H}$
 fields in $\hat{A} \rightsquigarrow$ bdd. lin map $\mathbb{H} \rightarrow \mathbb{H}$



in $\mathbb{C}$ ✓

in $\mathbb{D}$ ✓

in $\hat{A}$ bdd lin map $\mathbb{H} \rightarrow \mathbb{H}$

unitary VOA ✓

lin map on $\mathbb{D} \rightarrow \mathbb{D}$

$$\mathcal{D} := \text{span} \left\{ \begin{array}{l} A v_1(z_1) \dots v_n(z_n) \Omega \\ \text{for } v_i \in V, |z_i| < 1 \\ A \in \hat{A}^m \end{array} \right\} \quad \text{CH dense}$$

let $A \in \text{Univ}$

$$A = \mathbb{D} \setminus f(\hat{\mathbb{D}})$$

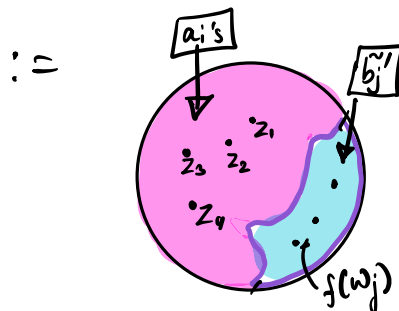
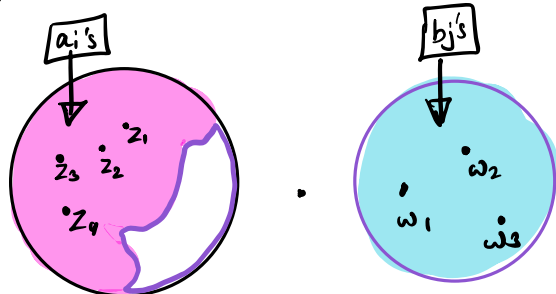
$$z_1, \dots, z_n \in \hat{A} \quad z_i \neq z_j$$

$$a_1, \dots, a_n \in V$$

$$A[a_1(z_1) \dots a_n(z_n)] : \mathcal{D} \longrightarrow \mathcal{D} \quad - \star$$

$$b_1(w_1) \dots b_m(w_m) \mapsto a_1(z_1) \dots a_n(z_n) \tilde{b}_1 f(w_1) \dots \tilde{b}_m f(w_m) \Omega \in \mathcal{D}$$

where $\tilde{b}_i$ is the image of b_i under change of coordinates induced by $f: (z_i) \mapsto f(z_i)$

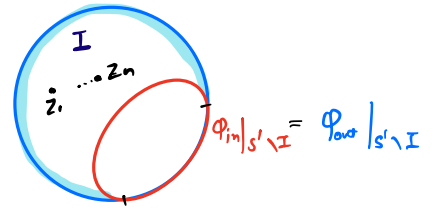


if VOA U is st $\star$ are bounded

$$A(I) := \cup N \left\{ A[\underline{a_1(z_1)} \dots \underline{a_n(z_n)}] \mid (A, \varphi_{in}, \varphi_{out}) \in \text{Ann} \right. \\ \left. \varphi_{in}|_{S' \setminus I} = \varphi_{out}|_{S' \setminus I} \right\}$$

define a cNet.

$a_i \in V \quad z_i \in \mathring{A}$



Th : (H-Tener)

Every cNet has an associated VOA.

A unitary VOA comes from a cNet iff * are bounded.

In fact, it is enough to check

$$V \longrightarrow H \quad \left. \vphantom{V \longrightarrow H} \right\} \text{ extends to map } H \rightarrow H$$

$$b \longmapsto a(z) f \cdot b$$

is bounded $\forall f: \mathbb{D} \rightarrow \mathbb{D}$ and $z \in \mathring{\mathbb{D}} \setminus f(\mathbb{D})$
 $0 \mapsto 0$

Conjecture all unitary VOA satisfy the condition

Th:

Part 1

A crash course on Von Neumann Algebras

let $\mathcal{H}$ be a Hilbert space. $\mathcal{B}(\mathcal{H})$ is all bounded linear operators on $\mathcal{H}$.

$$\|a\xi\|_{\mathcal{H}} \leq K \|\xi\|_{\mathcal{H}} \quad \forall \xi \in \mathcal{H} \quad \text{where} \quad \|\xi\| := \sqrt{\langle \xi, \xi \rangle}$$

- vector space
- Banach space (complete normed vector space) all Cauchy sequences converge
 $\|a\| := \sup \{ \|a\xi\| : \xi \in \mathcal{H}, \|\xi\| \leq 1 \}$
- Banach * algebra (operator product and adjoint $^+$ or *)

Let $A \subseteq \mathcal{B}(\mathcal{H})$ be a $*$ -subalgebra. Its commutant A' is defined as $\{ b \in \mathcal{B}(\mathcal{H}) \mid ab = ba \quad \forall a \in A \}$.

Clearly, $A \subset A''$

def: A is a von Neumann algebra if $A = A''$.

Example: let G be a group equipped with a faithful unitary representation $\alpha: G \rightarrow U(\mathcal{H})$

The group von Neumann algebra of the representation α is $\alpha(G)''$
 (can take $\ell^2(G)$ and action by left multiplication)

there are several topologies on $\mathcal{B}(\mathcal{H})$.

hardest to converge in

norm

$\Rightarrow$ convergence

$\supset$ ultra strong

strong

(like sand) fine

$\Rightarrow$

easiest to converge in (most # converging nets)

$\supset$ ultra weak

$\supset$ weak

coarse

$$A = A'' \Leftrightarrow A \text{ is weak closed.}$$

strong
ultra weak
ultra strong

{ has all limits of corresponding converging nets }

weak closed $\Rightarrow$ strong closed

$\Downarrow$
ultra weak closed

$\Rightarrow$ ultra strong closed

$\Downarrow$
norm closed

a (left) A -module is normal (cont. for ultraweak topology) $\neq$ homomorphism -
 $A-B(H)$ for a separable hilbert space H .

Defn. an A -module H is called finite if every A -linear isometry $H \rightarrow H$ is unitary.

$$\left\{ \begin{array}{l} u: H \rightarrow H \\ \text{s.t. } u^*u = I_H \end{array} \right. \begin{array}{l} \text{isometric} \\ \text{isomorphism.} \end{array} \quad A\text{-Mod}^f := \{(\text{finite } A\text{-modules})/\text{iso}, \oplus\}$$

$\downarrow$
 $u u^* = I_H$

If A is a factor, it has only one infinite module upto isomorphism.

Type classification

A is a factor if the centre is trivial

$$A \cap A' = \mathbb{C}$$

$$A\text{-Mod}^f \cong A\text{-Mod}$$

$$\text{I } (\mathbb{N}, +) \cup \{\infty\}$$

$$\text{II } (\mathbb{R}_{\geq 0}, +) \cup \{\infty\}$$

$$\text{III}_\lambda \quad \{0\} \cup \{\infty\}$$

$$\text{I}_n \quad \{0, 1, \dots, n\}$$

$$\text{I}_\infty \quad \mathbb{N} \cup \{\infty\}$$

$$\text{II}_1 \quad [0, 1]$$

$$\text{II}_\infty \quad \mathbb{R}_{\geq 0} \cup \{\infty\} \\ = \text{II}_1 \otimes \text{I}_\infty$$

$$\{0\} \cup \{\infty\}$$

$\text{Proj}(A)/\sim$
 $\{K \in A\text{-Mod} \mid K \leq H\}$
 initial segment of $A\text{-Mod}$

$B(H)$ for
 some hilbert
 space H .

trace class.
 'finite factors'

A state is a linear functional $\varphi: A \rightarrow \mathbb{C}$ that sends positive elements to positive elements. and $\varphi(a^*) = \overline{\varphi(a)}$.

$$a = x^2 \times \text{for some } x$$

$$:= A_+$$

A trace is a state that satisfies $\varphi(ab) = \varphi(ba) \quad \forall a, b \in A$

A state π faithful if $\varphi(a) = 0 \Rightarrow a = 0 \quad \forall a \in A$

Type I and II admit a base.

Given A and $A \trianglelefteq H$, $\Omega \in H$

Ω is cyclic if $A\Omega$ is dense in H

Ω is separating if $a \mapsto a\Omega$ is injective.

for a state ω , $\exists$
 $P_\omega \in A$ positive
 and self adjoint

$$\text{st } w(a) = \text{tr}(a P_w)$$

$$S_{\psi} = -\text{tr}(\rho_{\psi} \log \rho_{\psi})$$

QNS

$$H, \Omega \mapsto \langle -, \Omega, \Omega \rangle_H$$

$$\{\text{representation } A \hookrightarrow H + \underbrace{\Omega \in H}_{\text{cyclic}}\} \iff \{\text{state } \varphi : A \rightarrow \mathbb{C}\}$$

defines an inner product on A by $\langle a, b \rangle_\varphi := \varphi(b^*a)$

$L^2(A, \varphi)$ is completion of A w.r.t the norm induced by this inner product.

$\Omega \in L^2(A, \varphi)$ is the image of $1 \in A$

Tomita-Takesaki theory

for an arbitrary von Neumann algebra A , a faithful state

$$\varphi: A \rightarrow \mathbb{C}$$

$$L^2 A := L^2(A, \varphi)$$

$$\left\{ \begin{array}{l} \text{for two states } \varphi_1 \text{ and } \varphi_2 \\ L^2(A, \varphi_1) \cong L^2(A, \varphi_2) \end{array} \right\}$$

canonical unitary

$$A \subset L^2 A$$

$$a \mapsto a \cdot \Omega \in L^2 A$$

1. Consider $S_0: x \cdot \Omega \mapsto x^* \cdot \Omega$
and S be its closure.

2. $\Delta := S^* S$

$$S = J \Delta^{1/2} \quad (J := \text{closure of } S \Delta^{-1/2})$$

* $J A J = A'$

$$\Delta^{-it} A \Delta^{it} = A \quad \forall t \in \mathbb{R} \quad \leftarrow \text{action of } \mathbb{R} \text{ on } A$$

for A a type III_1 factor,

$$\hat{A}_{\text{ext}} \hookrightarrow \mathbb{H} \otimes L^2 \mathbb{R}$$

generated by

$$\begin{array}{c} a \otimes 1 \\ \Delta^{-it} \otimes e^{ixt} \end{array}$$